(*c*) Population and sampling

Students should be taught to:

understand the meaning of the term population;	The definition of 'population' can vary – eg it could be a family or the cars in a car park.
understand the word census, especially with regard to well defined, small scale and large populations, eg National census;	A census obtains information about every member of a population. **The types of questions used and how the collected data is used.**
understand the reasons for sampling and that sample data is used to estimate values in the population;	Reasons to include time and efficiency, and the impossibility of reaching the whole population in many circumstances.
understand the terms random, randomness and random sample;	The relation between 'random' and 'equally likely' may be tested.
generate and use random numbers using a random number table, calculator or computer including the use of a spreadsheet;	
understand, design and use a sampling frame;	Designing a sampling frame might be tested.
be able to select a simple random sample or a stratified sample by **more than** one category as a method of investigating a population;	An appreciation of an appropriate sample size will be expected, as will the ability to make a random selection or sample from a population using tables of random numbers or a calculator. Examples of one category might include male/female or KS2/KS3/KS4.
have a basic idea of the concept of bias, how it might occur in a sampling procedure and how it might be minimised;	Possible bias in sources of secondary data, eg vested interests.
understand and use systematic, quota and cluster sampling;	**With particular reference to large scale lines of enquiry such as quality control or opinion polls.**
understand the strengths and weaknesses of various sampling methods, including bias, influences and convenience.	**An awareness of influences such as gender, social background or geographical area will be expected.**